

Aynaz Javanivayeghan

AI / Machine Learning Engineer | Software Developer | Data Scientist

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Summary

MSc Computer Science candidate at Concordia University (GPA 3.80/4.0) with hands-on experience training and deploying deep learning models, LLM applications, and end-to-end ML pipelines. Co-founded an LLM optimization platform serving real-time inference; currently building large-scale time-series models for power-grid forecasting. Seeking internship or entry-level roles in AI/ML Engineering, Software Development, or Data Science.

Experience

Graduate Research Assistant (Volt-Age)

May 2025 – Present

- Design and deploy real-time deep learning models for power-grid outage forecasting using climate, telemetry, and historical outage data to improve grid resilience.
- Engineer scalable time-series pipelines that ingest, clean, and align multi-source environmental and infrastructure data for downstream model training.
- Collaborate with a multi-disciplinary research team on large-scale AI system design, model evaluation, and reproducible experiment tracking.

Teaching Assistant, Computer Science Department (Concordia University)

Sept 2024 – Present

- Support instruction, grading, and mentoring for 100+ students across 10+ undergraduate and graduate CS courses including Deep Learning, Intelligent Systems, Databases, OOP, Computer Organization, and Web Programming.
- Review student code, lead office hours, and develop instructional resources to strengthen practical programming and AI skills.

Python Developer & ML Intern (Bartarin Software Co.)

Jul 2022 – Feb 2023

- Built end-to-end ML pipelines covering data ingestion, preprocessing, model training, and evaluation for production use cases.
- Improved model prediction accuracy by 12% via systematic hyperparameter tuning, feature engineering, and cross-validation.
- Automated large-scale data-cleaning workflows in Python, reducing manual preprocessing effort and improving dataset quality for downstream models.

Technical Skills

- **Programming:** Python, Java, C, C++, JavaScript, SQL, HTML/CSS, Assembly, Bash
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, LangChain, LLMs, RAG, Prompt Engineering, OpenCV, SpeechBrain, Time-Series Forecasting, NLP, Computer Vision, Fine-Tuning, Transfer Learning
- **Data & Analytics:** Pandas, NumPy, SQL, Power BI, ETL Pipelines, Feature Engineering, A/B Testing, Statistical Analysis
- **Backend, Cloud & DevOps:** FastAPI, Flask, Django, REST APIs, AWS, Docker, Kubernetes, CI/CD, Git, GitLab, Linux, Firebase

Projects

41prompts.ai — Co-Founder & Engineering Lead

LLMs, LangChain, FastAPI, Docker, Kubernetes

- Led end-to-end development of a browser-based platform delivering real-time LLM prompt optimization, evaluation, and chaining.
- Architected a modular backend integrating multiple LLM providers behind a unified API for A/B testing and benchmarking.
- Deployed containerized microservices on Kubernetes for scalable, low-latency inference and continuous delivery.

Parkinson's Disease Detection from Speech

PyTorch, SpeechBrain, Hugging Face, MFCC, Transfer Learning

- Fine-tuned and benchmarked X-vector, ECAPA-TDNN, and Whisper models to classify Parkinson's Disease from voice recordings.
- Evaluated robustness across original and augmented datasets using MFCC features and transfer learning.

TubeLytics: YouTube Analytics System

Java, Akka, Play Framework, CI/CD

- Built a scalable analytics platform ingesting YouTube Data API streams via the actor model for high-throughput concurrent processing.
- Implemented unit tests and Git-hook CI to maintain code quality and prevent regressions.

Project Management: End-to-End PM Simulation

PMBOK, WBS, Gantt, EVM Fall 2025

- Drove a full-lifecycle project (initiation through closing) and produced core PM artifacts including WBS, Gantt chart, cost baseline, risk & quality registers, and EVM analysis.
- Delivered final reports and team presentation applying formal project-management methodology.

Education

Master of Science in Computer Science – GPA: 3.80/4.0

Concordia University

Montreal, Canada

Sept 2024 – Present

Thesis: Deep learning for power outage prediction using large-scale time-series and environmental data to improve grid reliability and outage response.

Relevant Coursework: Applied Artificial Intelligence, Machine Learning, Conversational AI, Advanced Programming, Project Management

Bachelor of Science in Computer Engineering – Azad University – GPA: 3.60/4.0

Sept 2018 – Jul 2022

Awards & Affiliations

Concordia Merit Scholarship (2024) — awarded for outstanding academic performance in the MSc Computer Science program.

GEMinAI Mentors & Mentees (2025 – Present) — Member of an AI-focused mentorship and professional-development community.